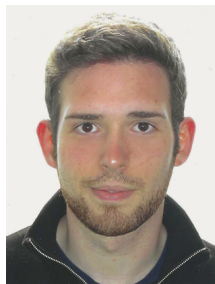


Curriculum Vitae of Dr. Néstor García-Romeral

(Updated 09/09/2026)



Place and date of birth: Barcelona, May 5th, 1998

Private address: Avinguda Diagonal 184, 6º 2ª, 08018 Barcelona, Spain

ORCID: 0000-0003-3129-3697

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Education

- **01/09/2022 – 15/07/2026:** PhD in Physical Chemistry with highest academic distinction (Excellent *Cum Laude* by unanimous committee vote), Universitat de Barcelona (Spain). Supervised by Prof. Francesc Illas and Prof. Ángel Morales-García. Thesis Title: “Modeling of MXene-Based Materials: From 2D Heterostructures to 0D Flakes”.
- **01/09/2020 – 15/07/2022:** Interuniversity Master in Theoretical Chemistry and Computational Modeling (120 ECTS), Universitat de Barcelona (Spain).
- **01/09/2016 – 15/07/2020:** Bachelor’s Degree in Chemistry, Universitat de Barcelona (Spain).

Present position

Postdoctoral Researcher in Physical Chemistry (15/07/2026 – Present)

Financed through competitive doctoral grant PREDOC-UB (2024), Universitat de Barcelona.

Focus on *ab initio* modelling of two-dimensional transition metal carbides/nitrides (MXenes) and Machine Learning Force Fields (MLFFs).

Departament de Ciència de Materials i Química Física

Facultat de Química, Universitat de Barcelona, C/ Martí Franquès 1, 08028 Barcelona, Spain.

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Professional Experience & University Positions

- **15/07/2026 – Present:** Postdoctoral Researcher in Physical Chemistry, Departament de Ciència de Materials i Química Física, Universitat de Barcelona (Spain). Financed through competitive grant PREDOC-UB (2024). Focus on *ab initio* modelling of two-dimensional transition metal carbides/nitrides (MXenes) and Machine Learning Force Fields (MLFFs).

- **01/11/2022 – 14/07/2026:** PhD Researcher, Departament de Ciència de Materials i Química Física, Universitat de Barcelona (Spain). Doctoral research funded through competitive fellowships: FISUR (01/11/2022 – 30/09/2025) awarded by AGAUR / Generalitat de Catalunya and PREDOC-UB (01/10/2025 – 15/07/2026) awarded by Universitat de Barcelona.
- **01/03/2022 – 30/06/2022:** Specialized Technician, Departament de Ciència de Materials i Química Física, Universitat de Barcelona (Spain). Project: “Rational design of new heterogeneous catalysts, electrocatalysts and photocatalysts for the production of sustainable clean energy through the conversion of H₂” (Code: RT2018-095460-B-100, funded by MCIU/AEI/FEDER, UE).
Tasks: Execution of high-performance quantum chemical and collinear calculations focused on magnetic coupling, electronic ground states, and magnetic exchange interactions in bare and functionalized 2D carbides (MXenes).
- **13/09/2021 – 02/11/2021:** Specialized Technician, Institut de Química Teòrica i Computacional (IQTUB), Universitat de Barcelona (Spain). Project: “María de Maeztu Unit of Excellence” (Code: MDM-2017-0767, funded by the State Research Agency - AEI).
Tasks: Study of the magnetic properties of two-dimensional MXene materials.
- **01/09/2020 – 31/08/2021:** Research Assistant, Fundació Bosch i Gimpera / IQTUB, Universitat de Barcelona (Spain). Project: “Advice and applied research in the field of nanomaterials modelling” (Code: 306478, led by Prof. Francesc Illas).
Tasks: First-principles calculations for low-dimensional materials and surface spectroscopy (XPS core-level shifts).
- **01/07/2019 – 31/07/2019:** Undergraduate Research Intern, Computational Materials Science Group (led by Prof. Dr. Francesc Illas), Universitat de Barcelona (Spain).
Tasks: Introductory research on quantum chemical modeling of functional nanomaterials, surface properties, and surface spectroscopy (XPS core-level shifts).

Research Stays & International Mobility

- **01/02/2024 – 01/05/2024 (3 months):** Visiting Scientist, Dipartimento di Scienza dei Materiali, Università degli Studi di Milano-Bicocca (Italy). Hosts: Prof. Gianfranco Pacchioni and Dr. Giovanni Di Liberto. Research Focus: First-principles modeling of supported oxide/MXene heterostructures, charge transfer mechanisms, and interface electronic structure.
- **21/11/2021 – 26/11/2021:** Course in Advanced Computational Techniques – Sorbonne Université (Paris, France). Master's program mobility.

- **24/10/2021 – 30/10/2021:** Course in Theoretical Methods for Simulation of Materials – Universidad Autónoma de Madrid & Instituto de Física Teórica (IFT-UAM/CSIC) (Madrid, Spain). Master's program mobility.
- **05/09/2021 – 18/09/2021:** Advanced Methods in Electronic Structure, Dynamics and Molecular Modeling – Université Toulouse-Jean Jaurès (Toulouse, France). Master's program mobility.

Specialized Courses & Training

- **03/09/2025 – 05/09/2025:** 1st Workshop of Initiation on Quantum Computing (12 hours). Organized by the Institut de Química Teòrica i Computacional (IQTCUB) and supported by the Consorci de Serveis Universitaris de Catalunya (CSUC).
- **2024 – 2025:** Machine Learning Specialization (DeepLearning.AI & Stanford University, Coursera). Tutored by Andrew Ng. Modules:
 - **19/08/2024 – 27/09/2024:** Supervised Machine Learning: Regression and Classification.
 - **30/09/2024 – 28/10/2024:** Advanced Learning Algorithms.
 - **02/12/2024 – 13/01/2025:** Unsupervised Learning, Recommenders, Reinforcement Learning.
- **25/06/2019 – 28/06/2019:** Introductory Course in Computational Chemistry. Institut de Química Teòrica i Computacional (IQTCUB), Universitat de Barcelona (Spain).

Main research interests

My main field is the application of *ab initio* computational methods to investigate and predict the physical properties of advanced solid-state materials. Driven by a deep interest in two-dimensional systems, my current focus areas include:

- **Electronic Structure of 2D Materials:** Property modulation and electronic characterization of transition metal carbides and nitrides (MXenes), encompassing pristine and functionalized layers, complex heterostructures, composites, and finite (0D) material models.
- **Solid-State Magnetism:** Modeling of magnetic exchange interactions, spin-state ordering, and electronic phenomena at the transition metal centers of functionalized 2D materials.
- **Surface Spectroscopy Modeling:** Computational modeling and interpretation of surface XPS core-level shifts to directly bridge theoretical predictions of surface chemistry with experimental characterizations.
- **Rational Design of Novel Phases:** Application of first-principles calculations to predict, tailor, and understand the fundamental properties of novel 2D electronic and magnetic phases for advanced applications.

- **Machine Learning Force Fields (MLFFs) for Nanostructures:** Development of training datasets, active learning strategies, and parameterization of ML-based potentials to enable large-scale dynamical simulations of finite 2D flakes and non-periodic nanostructures beyond standard DFT limits.

Publications

- García-Romeral, N.; Keyhanian, M.; Morales-García, Á.; Illas, F. Relating X-ray photoelectron spectroscopy data to chemical bonding in MXenes. *Nanoscale Adv.*, **2021**, 3, 2793–2801. <https://doi.org/10.1039/d0na01033b>. (Portada de revista / Journal Cover).
- García-Romeral, N.; Morales-García, Á.; Viñes, F.; Moreira, I. de P. R.; Illas, F. Theoretical Analysis of Magnetic Coupling in the Ti₂C Bare MXene. *J. Phys. Chem. C*, **2023**, 127, 3706–3714. <https://doi.org/10.1021/acs.jpcc.2c07609>.
- García-Romeral, N.; Morales-García, Á.; Viñes, F.; Moreira, I. de P. R.; Illas, F. How does thickness affect magnetic coupling in Ti-based MXenes. *Phys. Chem. Chem. Phys.*, **2023**, 25, 17116–17127. <https://doi.org/10.1039/D3CP01617J>. (Portada de revista / Journal Cover).
- García-Romeral, N.; Morales-García, Á.; Viñes, F.; Moreira, I. de P. R.; Illas, F. The nature of the electronic ground state of M₂C (M = Ti, Zr, Hf, V, Nb, Ta, Cr, Mo, and W) MXenes. *Phys. Chem. Chem. Phys.*, **2023**, 25, 31153–31164. <https://doi.org/10.1039/D3CP04402E>.
- García-Romeral, N.; Keyhanian, M.; Morales-García, Á.; Viñes, F.; Illas, F. Understanding the Chemical Bond in Semiconductor/MXene Composites: TiO₂ Clusters Anchored on the Ti₂C MXene Surface. *Chem. Eur. J.*, **2024**, 30, e202400255. <https://doi.org/10.1002/chem.202400255>.
- Keyhanian, M.*; García-Romeral, N.*; Morales-García, Á.; Viñes, F.; Illas, F. First principles modeling of composites involving TiO₂ clusters supported on M₂C MXenes. *Phys. Chem. Chem. Phys.*, **2024**, 26, 25319–25328. <https://doi.org/10.1039/D4CP01757A>. (Portada de revista / Journal Cover, *co-primer autor).
- García-Romeral, N.; Morales-García, Á.; Viñes, F. Effect of the Ti₂CT_x (T_x = O, OH, and H) Functionalization on the Formation of (TiO₂)₅/Ti₂CT_x Composites. *J. Phys. Chem. C*, **2024**, 128, 18450–18460. <https://doi.org/10.1021/acs.jpcc.4c06909>.
- García-Romeral, N.; Di Liberto, G.; Morales-García, Á.; Viñes, F.; Illas, F.; Pacchioni, G. Functionalization-Driven Formation of TiO₂/Ti₂CT_x Interfaces. *J. Phys. Chem. C*, **2025**, 129, 19551–19559. <https://doi.org/10.1021/acs.jpcc.4c08400>.

- García-Romeral, N.; Morales-García, Á.; Viñes, F.; Illas, F. Cutting edge(s): towards realistic modelling of MXene flakes. *Nanoscale*, **2025**, *17*, 10820–10831. <https://doi.org/10.1039/D5NR02231B>. (Portada de revista / Journal Cover).

Open-Source Scientific Software & Code Repositories

- **García-Romeral, N. Heterostructures-TiO₂-MXene (2024).** Open-source computational code developed in Python for the atomistic construction, geometric optimization, and interface matching of 0D oxide clusters supported on 2D MXene monolayers. Repository: [https://github.com/NestorGRG/heterostructures-TiO₂-MXene](https://github.com/NestorGRG/heterostructures-TiO2-MXene).

Contributions to Scientific Conferences & Workshops

- **28/04/2026:** Invited Departmental Seminar: “Computational Modelling of MXene-Based Materials: From 2D Heterostructures to 0D Flakes” – Technical University of Denmark (DTU Physics), Copenhagen, Denmark.
- **2025:** Poster presentation: “MXene Flakes: Beyond periodic models” – 13th Triennial Congress of the World Association of Theoretical and Computational Chemists (WATOC), Oslo, Norway.
- **2025:** Oral communication & poster: “Estudi Teòric de la Formació de Interfases d'Heteroestructures Tipus TiO₂/Ti₂CT_x” – 2nd Meeting of Theoretical and Computational Catalan Chemistry Society, Barcelona, Spain.
- **2024:** Conference organization & poster: Organizing Committee member & Poster presentation “Understanding the formation of Titania/MXene composites” – Twins in Catalysis: Merging Theory and Experiment (In honor of the 70th birthday of Profs. Francesc Illas and Gianfranco Pacchioni), Barcelona, Spain.
- **2024:** Oral communication: “Insights on the nature of the chemical bond of composites made of TiO₂ clusters supported on Ti₂C from density functional theory” – First EuroMXene Congress, Valencia, Spain.
- **2024:** Oral communication: “Understanding The Magnetic Nature of Bare MXenes in their Electronic Ground State” – Electronic Structure Principles and Applications and Biannual Meeting of the RSEQ Group in Chemistry and Computation (ESPA 2024), Tarragona, Spain.
- **2023:** Poster presentation: “How does thickness affect magnetic coupling on Ti-based MXenes” – XXXIX Biennial Meeting of the Spanish Royal Society of Chemistry (RSEQ), Zaragoza, Spain.

- **2023:** Poster presentation: “A Theoretical Analysis Of Magnetic Coupling In MXene Ti_2C ” – 1st Meeting of Theoretical and Computational Chemistry, Catalan Chemistry Society, Barcelona, Spain.
- **2021:** Oral communication: “Relating XPS data to chemical bonding in MXenes” – Colloid and interface research & innovations Workshop (“EXTREME” project), Varna, Bulgaria. “EXTREME” project has received funding from the Ministry of Education and Science, National Program “European Scientific Networks”.

Relevant national and international collaborations

Active

- Prof. Michel W. Barsoum (Drexel University, USA) & Dr. Mihaela Florea (National Institute of Materials Physics, Romania): Experimental synthesis, characterization, and direct conversion of CH_4 to formaldehyde using MXene-derived oxide catalysts at low temperature.
- Prof. Narcís Homs & Prof. Pilar Ramírez de la Piscina (Universitat de Barcelona, Spain): Joint computational-experimental reaction pathway mapping and vibrational spectroscopy for CO_2 activation in the Reverse Water Gas Shift (RWGS) reaction in MAX phases and oxotitanates.
- Dr. Lorenzo Mino (Università degli Studi di Torino, Italy): Computational modelling and experimental surface spectroscopic signatures (XPS, vibrational analysis) of MAX phases and oxotitanates.

Past few years

- Prof. Gianfranco Pacchioni & Dr. Giovanni Di Liberto (Università degli Studi di Milano-Bicocca, Italy): First-principles modeling of supported $\text{TiO}_2/\text{MXene}$ 2D/2D heterostructures, interfacial charge transfer, and electronic structure modulation.
- Dr. Masoomeh Keyhanian (Sofia University, Bulgaria): First-principles study and structural descriptor development for TiO_2 cluster composites supported on M_2C MXene surfaces.
- Prof. Ibério de P. R. Moreira (Universitat de Barcelona, Spain): Theoretical analysis of magnetic exchange coupling, spin-state ordering, and open-shell ground states in low-dimensional transition-metal carbides.

Teaching & Mentoring Experience

- **Teaching Assistant:** Computer Laboratories in Physical Chemistry II, Chemistry Degree, Universitat de Barcelona (hands-on lectures on Quantum Chemistry applications).

- **Teaching Assistant:** Computer Resources for Chemistry Degree, Universitat de Barcelona.
- **Student Mentoring:** Informal direct co-supervision of undergraduate research projects and software onboarding for junior PhD candidates in computational chemistry.

Competitive Funding & Grants Received

- **PREDOC-UB Doctoral Grant (2025 – 2026):** Competitive doctoral grant awarded by the Universitat de Barcelona to complete PhD research and postdoctoral bridge research.
- **FI-SDUR Doctoral Grant (2022 – 2025):** Competitive PhD fellowship awarded by AGAUR (Generalitat de Catalunya) / European Social Fund to support doctoral training in Physical Chemistry.
- **IQTCUB Master's Grant (2020 – 2021):** Competitive initiation-to-research fellowship awarded by the Institute of Theoretical and Computational Chemistry of the Universitat de Barcelona (IQTCUB) for Master's thesis research under the María de Maeztu Unit of Excellence program (MDM-2017-0767).
- **General Master's Fellowship (2020 – 2022):** Competitive general scholarship awarded by the Spanish Ministry of Education for university master's studies.
- **Ministry Collaboration Scholarship (2019 – 2020):** Competitive university collaboration grant awarded by the Spanish Ministry of Education. Research project: “Analysis of the Chemical Bonding on Carbide MXenes from the calculated C(1s) Core Level Binding Energy” (Conducted at IQTCUB).
- **General Undergraduate Scholarship (2016 – 2020):** Competitive scholarship awarded by the Spanish Ministry of Education covering the entire Bachelor's Degree in Chemistry.

Languages

- **Catalan:** Native.
- **Spanish:** Native.
- **English:** Professional Working Proficiency (Level B2, Cambridge English: First Certificate in English – FCE, awarded 2020).